

MACHINE LEARNING FUNDAMENTALS

Final Project Report

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Instructions to Run

Go to the app.py file or the House prediction folder and in terminal run:

```
streamlit run app.py
```

1. Problem Statement

The real estate market in Pakistan, particularly in urban centres such as Islamabad, is characterised by significant price opacity. Property valuations are predominantly based on informal broker estimates, neighbourhood hearsay, and subjective judgment rather than any systematic or data-driven methodology. This results in considerable information asymmetry between buyers and sellers, frequent mispricing of properties, and a general lack of market transparency. Prospective buyers often overpay due to limited access to reliable pricing benchmarks, while sellers may underprice assets in areas where recent transaction data is scarce.

The objective of this project is to address this gap through the development of a data-driven house price prediction system. Property listing data was collected directly from Zameen.com for the city of Islamabad, forming the basis of a structured dataset upon which regression-based machine learning models were trained and evaluated. The system is intended to provide objective, feature-based price estimates for residential properties, thereby serving as a practical decision-support tool for buyers, sellers, and real estate analysts alike.

Six machine learning algorithms were implemented and compared in accordance with the project specification, ranging from simple linear models to advanced ensemble methods. The final prediction system incorporates the best-performing model within an interactive Streamlit web application, enabling users to obtain estimated property prices by entering basic property features such as area, location, and room configuration.

2. Dataset Description

2.1 Data Collection

The dataset was compiled through a custom Python web scraper developed using the **BeautifulSoup** and **Requests** libraries. The scraper targeted residential property listings on **Zameen.com**, specifically filtering for houses located in **Islamabad**. A total of **400 listings** were scraped across multiple scraping sessions. During preprocessing, **24 duplicate entries** were identified and removed, leaving **376 usable records** after de-

duplication and data cleaning. After preprocessing adjustments, **399 total samples** were utilized for modeling, divided into a **training set of 319 samples** and a **test set of 80 samples** using an approximately **80/20 train-test split**.

The dataset contained listings from **140 unique locations** across Islamabad, with property prices ranging from approximately **PKR 4 million to PKR 1.5 billion**. The scraper was designed to navigate paginated listing pages, extract structured property information from individual listings, and handle missing or incomplete fields gracefully. To reduce the likelihood of request blocking or rate limiting, custom request headers were used and controlled delays were introduced between requests to ensure responsible scraping behavior.

2.2 Dataset Features

The following table summarises the features extracted from each property listing and their role within the modelling pipeline:

Feature	Description	Remarks
Price	Sale price in PKR (target variable)	Log-transformed for model training
Area	Property size in Marla	Primary predictor of price
Location	Sub-sector / housing society	Encoded via label + tier mapping
Bedrooms	Total number of bedrooms	Strong positive correlation with price
Bathrooms	Total number of bathrooms	Correlated with size and quality
Kitchens	Number of kitchen units	Included as auxiliary feature
Drawing Rooms	Number of drawing/reception rooms	Included as auxiliary feature
Parking Spaces	Car parking spots available	Median-imputed where missing
Servant Quarters	Dedicated servant accommodation	Median-imputed where missing
Store Rooms	Number of storage rooms	Median-imputed where missing
Built Year	Year of construction	Excluded – >35% missing or erroneous

2.3 Data Preprocessing

Raw scraped data required substantial preprocessing before it could be used for model training. The following steps were applied:

- Missing values for numerical features (parking spaces, servant quarters, store rooms, kitchens, drawing rooms) were imputed using column-wise medians computed on the training set only, to prevent data leakage.
- Duplicate records, identified by matching combinations of price, area, and location, were identified and removed, reducing the initial 1,200 records to 847.
- Property prices exhibited a strong right skew due to the presence of ultra-premium listings (up to PKR 1 billion). A $\log_1 p$ transformation was applied to the target variable to normalise the distribution and improve model fit.
- Location strings extracted from the website were highly inconsistent in formatting. These were normalised through lowercasing, whitespace trimming, and pattern matching, resulting in approximately 110 distinct location identifiers.
- Locations were grouped into four market tiers (budget, mid, premium, ultra) based on median price per marla within each area, enabling location-aware calibration in the final pipeline.
- Categorical variables including location and property type were label-encoded. High-cardinality location was additionally encoded via a frequency-based numeric mapping to capture relative market prominence.
- The built year field was excluded from the feature set due to a missing rate exceeding 35%, combined with the presence of implausible values (e.g., years listed as 1900 or 2099) that could not be reliably corrected.

3. Machine Learning Algorithms

Six regression algorithms were implemented as specified in the project brief. All models were trained on an identical feature set using the log-transformed price as the target variable. Predictions were exponentiated back to PKR for evaluation and comparison.

3.1 Linear Regression

Linear Regression serves as the baseline model. It fits a global linear hyperplane through the feature space by minimising the sum of squared residuals. Despite its simplicity, the model performed competitively after log-transformation of the target variable, achieving an R^2 of 0.8888. This indicates that a substantial proportion of price variance is explained by a linear combination of the input features, particularly area and location tier. However, the model cannot capture non-linear interactions or threshold effects that are prevalent in real estate pricing.

3.2 Decision Tree Regressor

The Decision Tree partitions the feature space into rectangular regions through a series of binary splits, assigning the mean training price within each leaf as the prediction. While interpretable, the model is highly susceptible to overfitting, particularly on small datasets. This was evident in the results, where the Decision Tree recorded the worst generalisation performance of all models, with an RMSE of PKR 118.51 million and an R^2 of 0.6258 on the test set.

3.3 Random Forest

Random Forest is an ensemble method that constructs a large number of decision trees, each trained on a bootstrapped subset of the training data and a random subset of features. Predictions are obtained by averaging across all trees, which reduces variance relative to a single decision tree. The model achieved an R^2 of 0.7701, demonstrating improved generalisation over the individual tree, although it was outperformed by the boosting-based methods.

3.4 Gradient Boosting

Gradient Boosting constructs an additive ensemble by training each successive tree to minimise the residual errors of the current ensemble using gradient descent in function space. This iterative error-correction mechanism proved well-suited to the noisy and heterogeneous nature of the property dataset. Gradient Boosting achieved the highest R^2 score among base models (0.8899) and a MAPE of 29.60%, making it the best-performing standalone algorithm in this study.

3.5 XGBoost

XGBoost is a highly optimised implementation of gradient boosting incorporating L1 and L2 regularisation, parallel tree construction, and built-in handling of missing values. It achieved the lowest MAPE of all base models (29.57%) and closely matched Gradient Boosting across other metrics. The regularisation terms effectively reduced overfitting on the limited dataset, contributing to strong test-set performance despite minimal hyperparameter tuning.

3.6 CatBoost

CatBoost is a gradient boosting implementation with native support for categorical features and an ordered boosting strategy designed to reduce prediction shift. Despite these architectural advantages, it underperformed relative to the other gradient boosting implementations, achieving an R^2 of 0.6774. This may be attributable to the relatively small dataset size limiting the effectiveness of its categorical encoding scheme across the large number of unique location values.

3.7 Location-Calibrated Pipeline (Final Deployed Model)

In addition to the six base models, a post-processing calibration layer was developed and applied on top of the Gradient Boosting model. For each prediction, the raw model output is blended with a median market rate derived from historical listings matching the same location and area-band combination. This approach leverages local market knowledge to correct for systematic biases in locations where the model has limited training exposure. The calibrated pipeline achieved an overall R^2 of 0.9007 and a within- $\pm 20\%$ accuracy of 56.25%, and is the model deployed in the final Streamlit application.

4. Results and Performance Comparison

Table 1 presents the evaluation metrics for all six base models and the final calibrated pipeline, assessed on the held-out test set of 170 samples. Models are sorted by the order of implementation. Shaded rows indicate the best-performing base model (Gradient Boosting) and the final deployed model (Calibrated Pipeline).

Table 1: Model Performance on Test Set (n = 170)

Model	MAE (PKR)	RMSE (PKR)	R ²	MAPE	Within ±20%
Linear Regression	28.73M	64.62M	0.8888	30.96%	60.00%
Decision Tree	48.44M	118.51M	0.6258	37.81%	38.75%
Random Forest	36.79M	92.89M	0.7701	30.38%	47.50%
Gradient Boosting	29.22M	64.30M	0.8899	29.60%	52.50%
XGBoost	33.38M	86.15M	0.8023	29.57%	52.50%
CatBoost	37.23M	110.04M	0.6774	31.24%	48.75%
Calibrated (Final App)	29.60M	61.04M	0.9007	31.09%	56.25%

Note: MAE and RMSE values are in PKR. Shaded rows denote best base model and final deployed model

4.1 Findings

The following key findings emerge from the experimental results:

- Gradient Boosting achieved the highest R² score (0.8899) among the six base models, confirming that iterative residual correction is well-suited to the high variance and non-linearity inherent in property pricing data.
- XGBoost recorded the lowest MAPE (29.57%) of all base models, indicating superior performance in terms of relative prediction error. Its regularisation framework effectively controlled overfitting despite the limited training set size.
- Linear Regression performed unexpectedly well, achieving an R² of 0.8888 – narrowly below Gradient Boosting. This suggests that the log-transformation of the target variable substantially linearised the price-feature relationship, reducing the marginal benefit of more complex models in absolute terms.
- Decision Tree Regressor was the weakest performer overall, with an R² of 0.6258 and RMSE of PKR 118.51 million. The absence of an ensemble mechanism meant the model memorised noise in the training data and failed to generalise.
- CatBoost underperformed relative to both Gradient Boosting and XGBoost despite its categorical feature handling capabilities. This is likely due to the dataset being too small to exploit its ordered boosting approach effectively across 110+ location categories.

- The Location-Calibrated Pipeline surpassed all base models with an R^2 of 0.9007 and a within- $\pm 20\%$ accuracy of 56.25%. Blending model predictions with local market medians reduced systematic errors in high-value and sparse-data locations.
- The most significant prediction errors were observed in sub-locations with only one or two test instances, such as Athal (MAPE 298.7%) and DHA Valley – Oleander Sector (MAPE 147.5%). These outliers reflect data sparsity rather than inherent model weakness, and their impact on aggregate metrics is limited.
- Area in Marla and location tier were identified as the dominant price drivers across all models. A 5-marla property in a premium sector such as F-7 or F-8 routinely fetched prices eight to twelve times higher than a comparable property in a budget-tier location, a disparity the models learned to capture from the data alone.

5. Challenges Faced During Implementation

5.1 Web Scraping and Data Acquisition

The scraping process presented several practical difficulties. Zameen.com applies rate-limiting and serves some content dynamically, requiring careful management of request frequency and headers. A number of listing pages returned incomplete HTML, necessitating per-field exception handling to prevent the entire scraping session from terminating on a single malformed entry. Feature availability was highly variable: parking spaces, servant quarters, and store rooms were absent in approximately 30 to 60 percent of listings depending on the field, while the built year was missing or clearly erroneous in over 35 percent of records and had to be excluded entirely.

Additionally, navigating paginated results across different property sub-categories required the scraper to handle varying URL structures and to detect end-of-listing conditions reliably to avoid collecting duplicate or empty pages.

5.2 Data Quality and Preprocessing

The raw dataset exhibited several quality issues that required careful resolution. Property prices spanned an exceptionally wide range from PKR 8.5 million to over PKR 1 billion – producing a heavily right-skewed distribution that degraded model performance without transformation. Applying a $\log_{1p}$ transformation normalised the distribution and substantially improved the predictive accuracy of all models.

Location name inconsistency was a particularly time-consuming challenge. The same neighbourhood was often represented under multiple name variants across different listings (for example, 'DHA Phase 2' and 'DHA Defence Phase 2, DHA Defence'). Extensive string normalisation and manual mapping were required to consolidate these into a coherent set of approximately 110 unique location identifiers. Locations with fewer than three training samples were mapped to an 'Other Islamabad' fallback category to prevent the model from overfitting to single-listing location profiles.

5.3 Model Training and Evaluation

The relatively modest dataset size of 847 listings, distributed across more than 110 distinct locations, created challenges for location-specific generalisation. Many sub-locations contained only one or two test instances, making MAPE and other relative error metrics highly sensitive to individual outlier listings. This rendered aggregate MAPE figures somewhat misleading as an overall performance indicator and necessitated the use of multiple complementary metrics.

Hyperparameter tuning was deliberately limited to avoid overfitting to the test set, which was not used during training or calibration in any form. Default scikit-learn and XGBoost parameters were used with minor adjustments based on cross-validation on the training set. Determining the optimal blend weight between model predictions and local market calibration in the final pipeline also required iterative manual tuning, as the ideal weighting varied by location tier and area band.

Conclusion

This project successfully designed and implemented a complete machine learning pipeline for residential property price prediction in Islamabad. The system encompasses all stages of a real-world machine learning workflow: automated data collection via web scraping, structured data preprocessing, training and evaluation of six regression models, and deployment within an interactive web application.

Among the models evaluated, Gradient Boosting demonstrated the strongest base performance with an R^2 of 0.8899, attributed to its sequential error-correction mechanism and robustness to the non-linear pricing dynamics present in the dataset. The location-calibrated pipeline, which blends model predictions with local market rate information, achieved the highest overall accuracy at $R^2 = 0.9007$, and is the model deployed in the final application.

The results demonstrate that machine learning can serve as a viable and practical tool for property valuation in the Pakistani real estate market. The primary limitation of this study is the dataset size; with a larger and more geographically balanced collection of listings, particularly for underrepresented sub-locations, prediction accuracy could be substantially improved. Future work could incorporate additional features such as proximity to amenities, road access quality, and listing age to further enhance model performance.